

FedPrivTab：交互形式 + API 形式对比三个模型

GitHub：<https://github.com/oneflow1/FedPrivTab>

生成时间：2026-06-18 06:13:49 UTC。本 HTML 对照展示交互式产品流程、真实 API 脚本、matplotlib 绘图代码和对应 PNG 输出。

一、交互形式：产品精简版训练流程

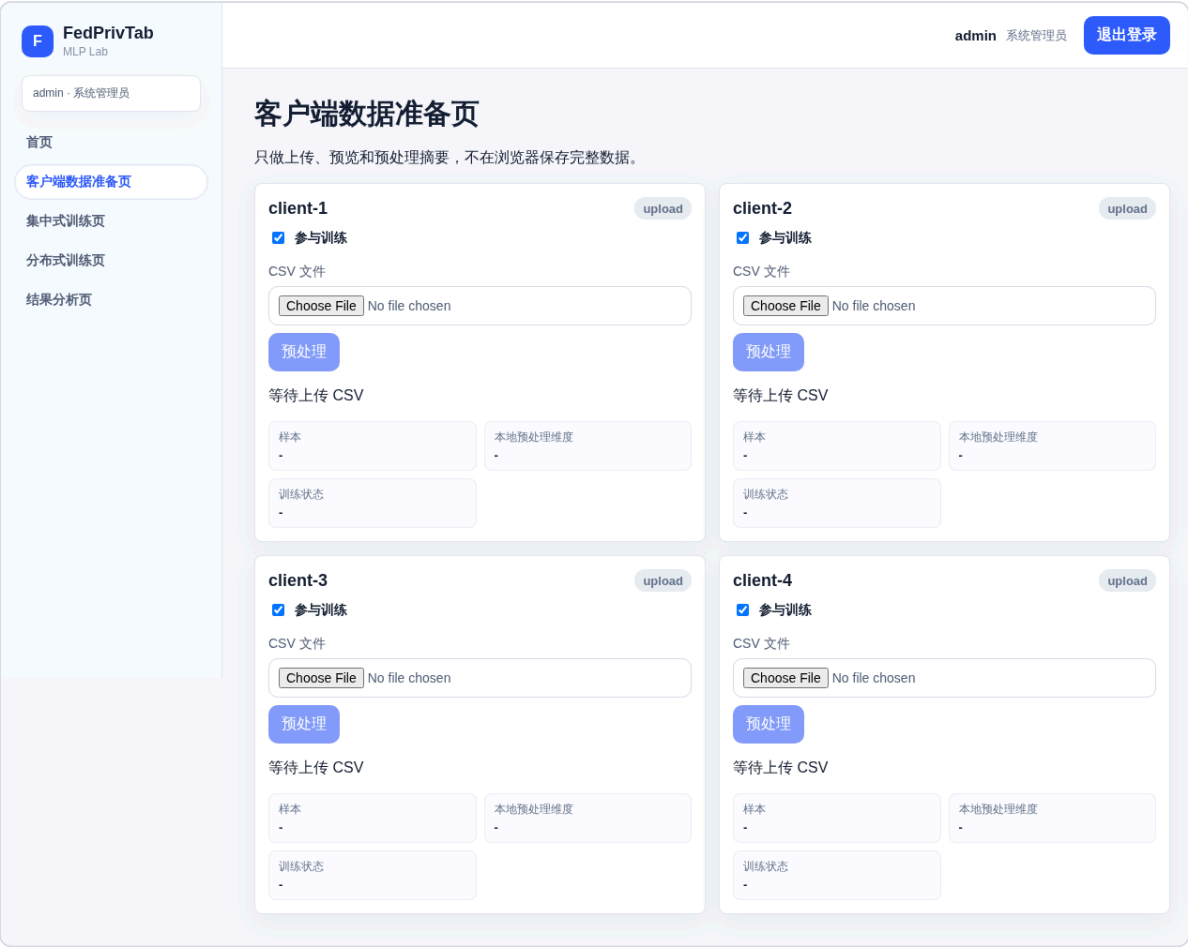
这一部分按 README 风格概览 FedPrivTab 的产品化训练路径：先登录进入工作台，再完成客户端数据准备，随后分别运行集中式 MLP、FedAvg MLP 和 DP-FedAvg MLP，最后在结果分析页查看指标与曲线。



图：登录页：输入账号后进入 FedPrivTab 产品工作台。



图：首页工作台：从导航入口进入数据准备、模型训练和结果分析流程。



图：客户端数据准备：上传或选择各客户端 CSV，完成联邦训练前的数据校验与预处理。

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集中式训练页

数据准备

CSV 文件

Choose File No file chosen

准备数据

等待上传 CSV

集中式 MLP

Epoch

Batch

学习率

隐藏层

激活函数

50

128

0.05

64,32

ReLU

开始集中式训练

图：集中式 MLP 训练：选择合并数据集并配置 epochs、学习率和 MLP 结构。

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分布式训练页

FedAvg 和 DP-FedAvg 只使用客户端数据准备页已预处理的数据；请先完成客户端预处理再启动训练。

可训练客户端

当前数据

联邦对齐维度

0

-

-

客户端卡片展示的是本地预处理维度；类别独热列会在后端联邦训练前按所有客户端的并集对齐，缺失特征填 0。

模式

FedAvg + MLP

Adult Non-IID 数据类别不均衡，推荐默认使用 50 轮和 step-decay 自适应学习率；轮数过少时早期结果可能停在多数类基线。

轮数

客户端数

本地 Epoch

学习率

Non-IID α

50

4

1

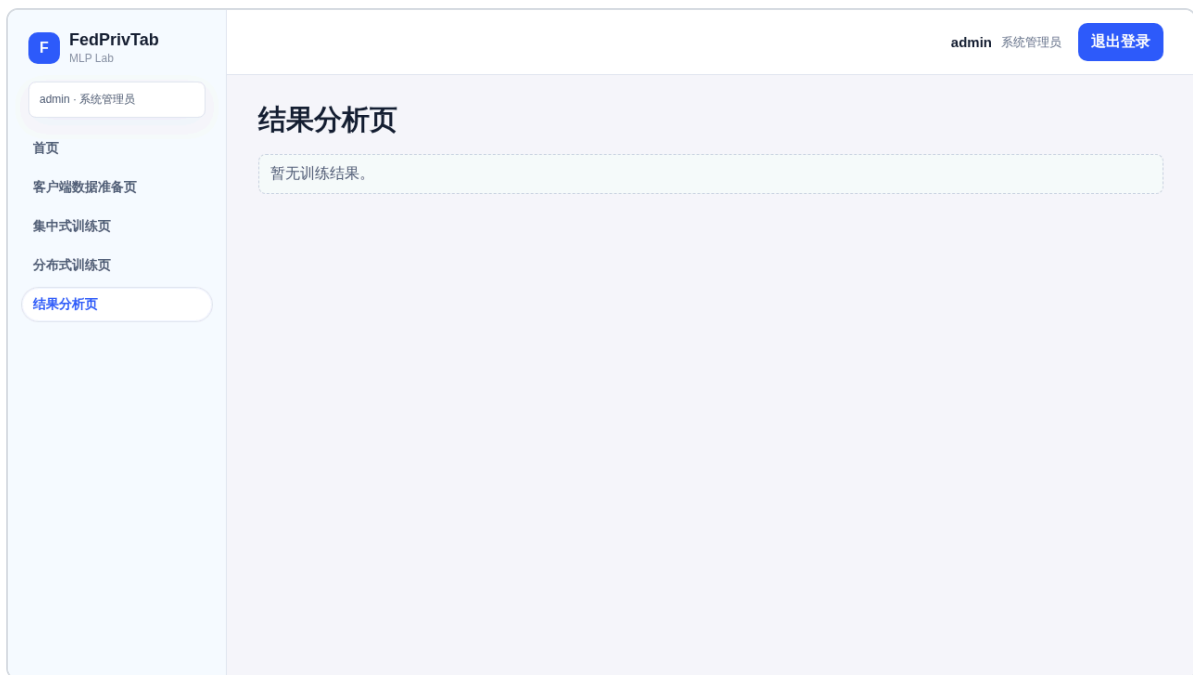
0.05

0.3

开始分布式训练

请先进入客户端数据准备页，完成客户端预处理后再启动分布式训练。

图：分布式 FedAvg / DP-FedAvg 训练：选择客户端数据集、通信轮次、客户端比例和 DP 参数。



图：结果分析：对比集中式 MLP、FedAvg MLP 与 DP-FedAvg MLP 的训练曲线和指标。

二、API 形式：真实接口训练脚本

API 执行模式：`http`。`experiment_results.json` 是 `/train` API 响应对象的缓存副本；当 `RUN_API_EXPERIMENTS=False` 时从缓存恢复 `api_results`，当该标志为 `True` 时同一个变量直接来自实时 `/train` 调用。`metrics_summary.csv` 只是由 API 结果派生出的辅助导出表，不作为绘图数据源。

默认配置来源

主实验默认配置由紧凑消融实验选择，消融记录保存于 `default_config_ablation.json`。集中式 MLP、FedAvg MLP 与 DP-FedAvg MLP 都使用相同 MLP 架构：`hidden_layers=2`、`hidden_units='64,32'`、`activation='ReLU'`。

```
import json
import time
from pathlib import Path

import requests

API_BASE = 'http://127.0.0.1:5000'
RUN_API_EXPERIMENTS = False
ROOT = Path('.')
OUT_DIR = ROOT / 'data/mywork/final_outputs'
RESULTS_JSON = OUT_DIR / 'experiment_results.json'
```

```

def post_preprocess(csv_path, client_id=None, role='centralized'):
    """Upload one CSV to /preprocess?async=true. role is 'client'/'centralized'."""
    csv_path = Path(csv_path)
    data = {'role': role}
    if client_id is not None:
        data['client_id'] = client_id
    with csv_path.open('rb') as fh:
        files = {'file': (csv_path.name, fh, 'text/csv')}
        response = requests.post(
            f'{API_BASE}/preprocess?async=true',
            files=files,
            data=data,
            timeout=60,
        )
    response.raise_for_status()
    return response.json()

def wait_job(job_id, poll_seconds=2, timeout_seconds=1800):
    """Poll /jobs/<job_id> until the async preprocessing job completes or fails."""
    deadline = time.time() + timeout_seconds
    while time.time() < deadline:
        response = requests.get(f'{API_BASE}/jobs/{job_id}', timeout=30)
        response.raise_for_status()
        payload = response.json()
        job = payload.get('job', payload)
        status = job.get('status')
        if status in {'completed', 'failed'}:
            if status == 'failed':
                raise RuntimeError(job)
            return job
        time.sleep(poll_seconds)
    raise TimeoutError(f'Job {job_id} did not finish within {timeout_seconds} seconds')

def completed_preprocess_result(start_response):
    job = start_response.get('job', start_response)
    job_id = job.get('job_id') or job.get('id')
    if job_id:
        job = wait_job(job_id)
        return job.get('result', job)
    return start_response

def post_train(payload):
    response = requests.post(f'{API_BASE}/train', json=payload, timeout=1800)
    response.raise_for_status()
    return response.json()

client_csvs = [
    ('client-1', ROOT / 'data/processed/adult_noniid_client_1.csv'),
    ('client-2', ROOT / 'data/processed/adult_noniid_client_2.csv'),
    ('client-3', ROOT / 'data/processed/adult_noniid_client_3.csv'),
    ('client-4', ROOT / 'data/processed/adult_noniid_client_4.csv'),
]

if RUN_API_EXPERIMENTS:
    central_preprocess = completed_preprocess_result(
        post_preprocess(ROOT / 'data/raw/adult.csv', role='centralized')
    )

```

```

    )
    client_preprocess = [
        completed_preprocess_result(post_preprocess(csv_path, client_id=client_id, role
        for client_id, csv_path in client_csvs
    ]
else:
    # experiment_results.json is a cached copy of previous /train API responses.
    # It is used only when RUN_API_EXPERIMENTS=False to avoid rerunning heavy training.
    saved = json.loads(RESULTS_JSON.read_text(encoding='utf-8'))
    central_preprocess = saved['metadata']['central_preprocess']
    client_preprocess = saved['metadata']['client_preprocess']

    central_dataset_id = central_preprocess['dataset_id']
    client_dataset_ids = [
        item['dataset_id']
        for item in client_preprocess
    ]
    client_datasets = [
        {'client_id': item['client_id'], 'dataset_id': item['dataset_id']}
        for item in client_preprocess
    ]

    centralized_payload = {
        'target_column': 'income',
        'batch_size': 128,
        'hidden_layers': 2,
        'hidden_units': '64,32',
        'activation': 'ReLU',
        'epochs': 50,
        'lr': 0.05,
        'lr_schedule': 'step_decay',
        'lr_decay': 0.5,
        'lr_step_size': 15,
        'lr_min': 0.005,
        'seed': 42,
        'mode': 'centralized',
        'dataset_id': central_dataset_id,
    }

    fedavg_payload = {
        'target_column': 'income',
        'clients': 4,
        'local_epochs': 1,
        'batch_size': 32,
        'hidden_layers': 2,
        'hidden_units': '64,32',
        'activation': 'ReLU',
        'rounds': 50,
        'lr': 0.05,
        'lr_schedule': 'step_decay',
        'lr_decay': 0.5,
        'lr_step_size': 15,
        'lr_min': 0.005,
        'seed': 42,
        'client_fraction': 1.0,
        'mode': 'fedavg',
        'dirichlet_alpha': 0.3,
        'client_dataset_ids': client_dataset_ids,
        'client_datasets': client_datasets,
    }

    dp_fedavg_payload = {

```

```

        **fedavg_payload,
        'mode': 'dp_fedavg',
        'noise_multiplier': 0.1,
        'lr': 0.03,
        'lr_schedule': 'step_decay',
        'lr_decay': 0.5,
        'lr_step_size': 15,
        'lr_min': 0.005,
        'epsilon': 4.0,
        'delta': 1e-5,
        'clip_norm': 1.0,
    }

    if RUN_API_EXPERIMENTS:
        api_results = [
            {'name': 'centralized_mlp', 'requested_config': centralized_payload, **post_train(fedavg_payload)},
            {'name': 'fedavg_mlp', 'requested_config': fedavg_payload, **post_train(fedavg_payload)},
            {'name': 'dp_fedavg_mlp', 'requested_config': dp_fedavg_payload, **post_train(dp_fedavg_payload)}
        ]
    else:
        # Same variable shape as live /train runs, loaded from the cached API response record
        api_results = saved['results']

    experiments = {item['name']: item for item in api_results}

    {
        'central_dataset_id': central_dataset_id,
        'client_dataset_ids': client_dataset_ids,
        'client_datasets': client_datasets,
        'train_payload_modes': [
            centralized_payload['mode'],
            fedavg_payload['mode'],
            dp_fedavg_payload['mode'],
        ],
        'api_results_source': 'live /train calls' if RUN_API_EXPERIMENTS else 'cached /train results',
        'loaded_existing_results': not RUN_API_EXPERIMENTS,
    }

```

主要指标

实验	模式	Accuracy	Precision	Recall	F1	AUC	噪声系数	alpha
centralized_mlp	centralized	0.8531	0.7268	0.6245	0.6718	0.9083		
fedavg_mlp	fedavg	0.8428	0.7332	0.5454	0.6255	0.8962		0.3000
dp_fedavg_mlp	dp_fedavg	0.8118	0.6877	0.4000	0.5058	0.8537	0.1000	0.3000
dp_noise_0.1	dp_fedavg	0.8118	0.6877	0.4000	0.5058	0.8537	0.1000	0.3000
dp_noise_0.2	dp_fedavg	0.7756	0.5400	0.4582	0.4957	0.7707	0.2000	0.3000
dp_noise_0.5	dp_fedavg	0.4850	0.3027	0.8740	0.4497	0.6681	0.5000	0.3000

实验	模式	Accuracy	Precision	Recall	F1	AUC	噪声系数	alpha
alpha_0.1	fedavg	0.8428	0.7332	0.5454	0.6255	0.8962		0.1000
alpha_0.3	fedavg	0.8428	0.7332	0.5454	0.6255	0.8962		0.3000
alpha_1.0	fedavg	0.8428	0.7332	0.5454	0.6255	0.8962		1.0000

alpha 敏感性实验使用显式客户端 CSV 固定划分，因此 alpha 主要是配置记录，不代表重新按 Dirichlet 分布切分后的客户端数据。

三模型 Accuracy/F1/AUC 对比

集中式 MLP、FedAvg MLP 与 DP-FedAvg MLP 的 Accuracy、F1 和 AUC 对比。

```
from pathlib import Path
import json
import numpy as np
import matplotlib.pyplot as plt

out = Path('data/mywork/final_outputs')
if not out.exists():
    out = Path.cwd()
# Plotting data comes from cached /train API response objects, not CSV summaries.
results = json.loads((out / 'experiment_results.json').read_text(encoding='utf-8'))
experiments = {item['name']: item for item in results['results']}

def metric_row(name):
    item = experiments[name]
    requested = item.get('requested_config') or {}
    dp = item.get('dp') or {}
    return {
        'name': name,
        **(item.get('metrics') or {}),
        'noise_multiplier': dp.get('noise_multiplier', requested.get('noise_multiplier')),
        'dirichlet_alpha': requested.get('dirichlet_alpha'),
    }

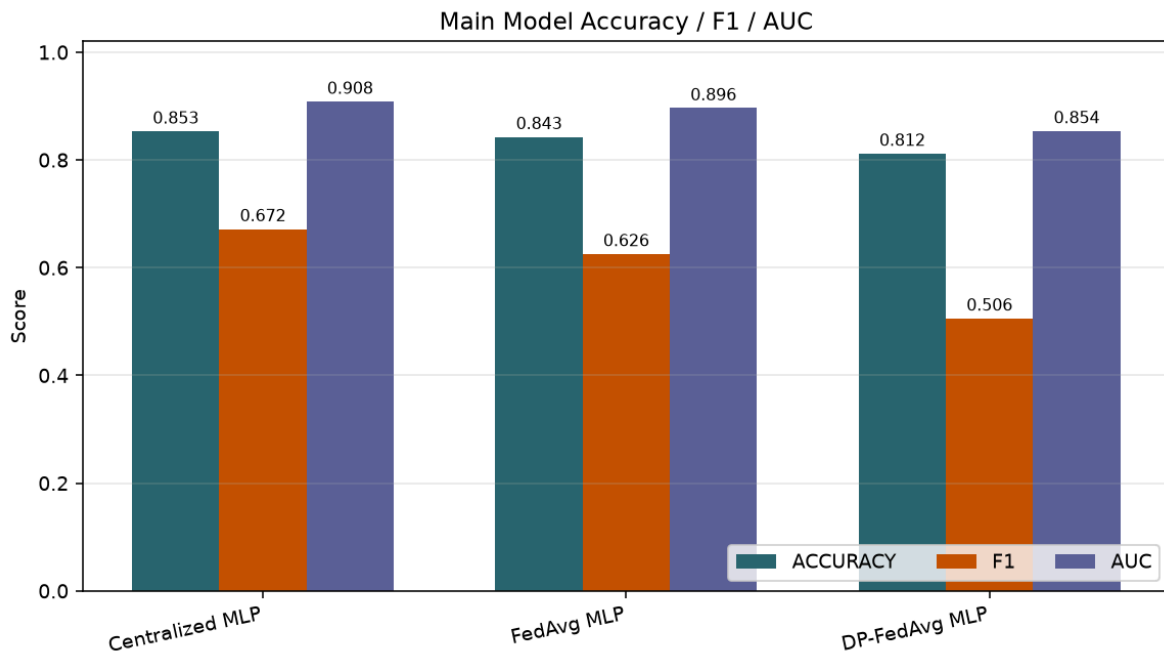
main_names = ['centralized_mlp', 'fedavg_mlp', 'dp_fedavg_mlp']
label_map = {
    'centralized_mlp': 'Centralized MLP',
    'fedavg_mlp': 'FedAvg MLP',
    'dp_fedavg_mlp': 'DP-FedAvg MLP',
}
metrics = ['accuracy', 'f1', 'auc']
plot_rows = [metric_row(name) for name in main_names]
x = np.arange(len(main_names))
width = 0.24
fig, ax = plt.subplots(figsize=(8.4, 4.8), dpi=140)
colors = ['#28666e', '#c75000', '#5b5f97']
for offset, metric, color in zip([-width, 0, width], metrics, colors):
```



```

values = [row[metric] for row in plot_rows]
bars = ax.bar(x + offset, values, width, label=metric.upper(), color=color)
ax.bar_label(bars, fmt='%.3f', fontsize=8, padding=2)
ax.set_xticks(x)
ax.set_xticklabels([label_map[name] for name in main_names], rotation=12, ha='right')
ax.set_ylim(0, 1.02)
ax.set_ylabel('Score')
ax.set_title('Main Model Accuracy / F1 / AUC')
ax.grid(axis='y', alpha=0.28)
ax.legend(ncols=3, loc='lower right')
fig.tight_layout()
plt.show()

```



metric_bar_accuracy_f1_auc.png

训练 Accuracy 曲线

三类主要训练流程的 Accuracy 随 epoch/round 的变化。

```

from pathlib import Path
import json
import numpy as np
import matplotlib.pyplot as plt

out = Path('data/mywork/final_outputs')
if not out.exists():
    out = Path.cwd()
# Plotting data comes from cached /train API response objects, not CSV summaries.
results = json.loads((out / 'experiment_results.json').read_text(encoding='utf-8'))
experiments = {item['name']: item for item in results['results']}

def metric_row(name):
    item = experiments[name]
    requested = item.get('requested_config') or {}

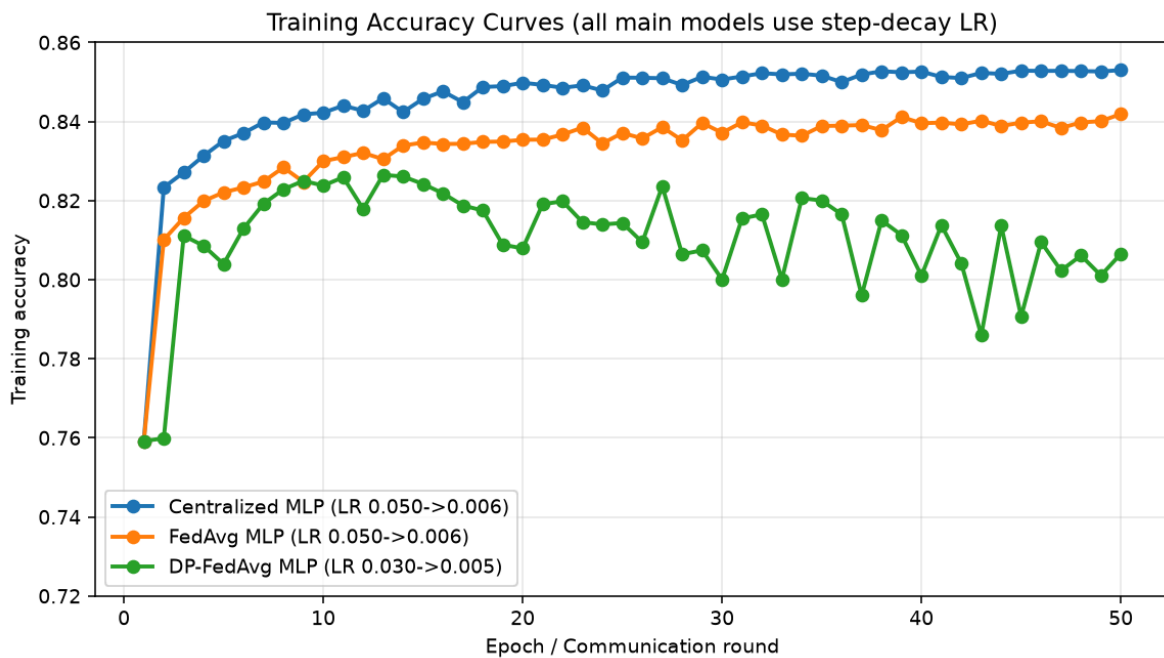
```

```

dp = item.get('dp') or {}
return {
    'name': name,
    **(item.get('metrics') or {}),
    'noise_multiplier': dp.get('noise_multiplier', requested.get('noise_multiplier')),
    'dirichlet_alpha': requested.get('dirichlet_alpha'),
}

main_names = ['centralized_mlp', 'fedavg_mlp', 'dp_fedavg_mlp']
label_map = {
    'centralized_mlp': 'Centralized MLP',
    'fedavg_mlp': 'FedAvg MLP',
    'dp_fedavg_mlp': 'DP-FedAvg MLP',
}
fig, ax = plt.subplots(figsize=(8.4, 4.8), dpi=140)
for name in main_names:
    values = experiments[name]['history']['accuracy']
    lr_values = experiments[name]['history'].get('lr', [])
    lr_note = ''
    if lr_values:
        lr_note = f" (LR {lr_values[0]:.3f}->{lr_values[-1]:.3f})"
    ax.plot(range(1, len(values) + 1), values, marker='o', linewidth=2, label=label_map[name] + lr_note)
ax.set_xlabel('Epoch / Communication round')
ax.set_ylabel('Training accuracy')
ax.set_title('Training Accuracy Curves (all main models use step-decay LR)')
ax.set_ylim(0.72, 0.86)
ax.grid(alpha=0.28)
ax.legend()
fig.tight_layout()
plt.show()

```



training_accuracy_curves.png

训练 Loss 曲线

三类主要训练流程的 Loss 随 epoch/round 的变化。

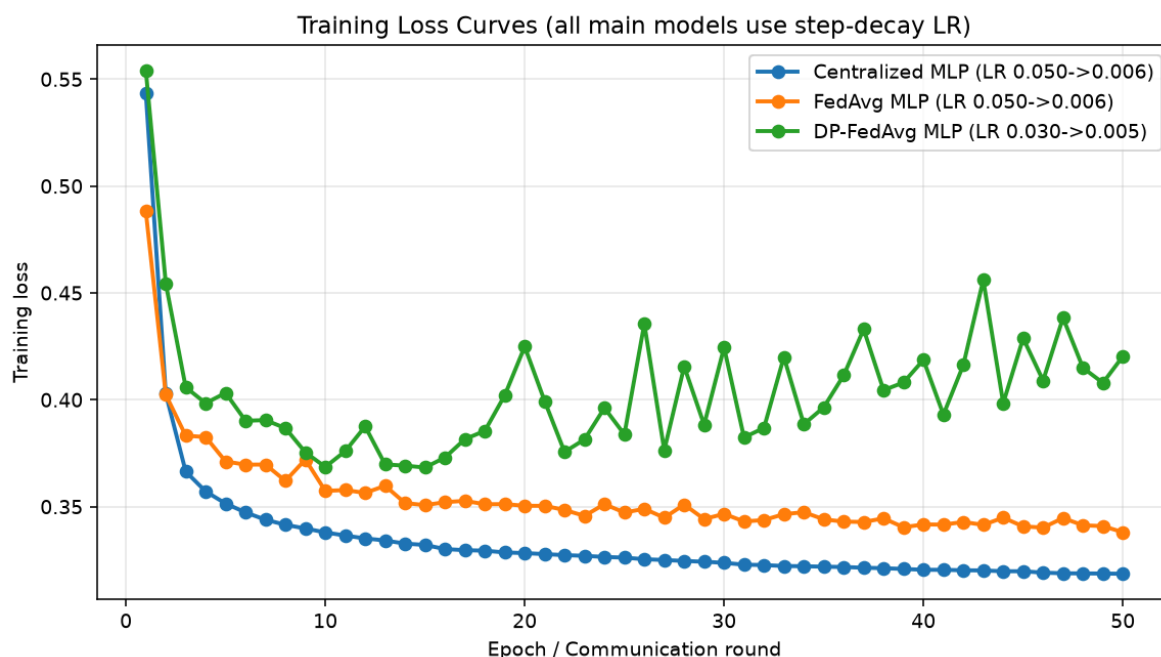
```
from pathlib import Path
import json
import numpy as np
import matplotlib.pyplot as plt

out = Path('data/mywork/final_outputs')
if not out.exists():
    out = Path.cwd()
# Plotting data comes from cached /train API response objects, not CSV summaries.
results = json.loads((out / 'experiment_results.json').read_text(encoding='utf-8'))
experiments = {item['name']: item for item in results['results']}

def metric_row(name):
    item = experiments[name]
    requested = item.get('requested_config') or {}
    dp = item.get('dp') or {}
    return {
        'name': name,
        **(item.get('metrics') or {}),
        'noise_multiplier': dp.get('noise_multiplier', requested.get('noise_multiplier')),
        'dirichlet_alpha': requested.get('dirichlet_alpha'),
    }

main_names = ['centralized_mlp', 'fedavg_mlp', 'dp_fedavg_mlp']
label_map = {
    'centralized_mlp': 'Centralized MLP',
    'fedavg_mlp': 'FedAvg MLP',
    'dp_fedavg_mlp': 'DP-FedAvg MLP',
}

fig, ax = plt.subplots(figsize=(8.4, 4.8), dpi=140)
for name in main_names:
    values = experiments[name]['history']['loss']
    lr_values = experiments[name]['history'].get('lr', [])
    lr_note = ''
    if lr_values:
        lr_note = f" (LR {lr_values[0]:.3f}->{lr_values[-1]:.3f})"
    ax.plot(range(1, len(values) + 1), values, marker='o', linewidth=2, label=label_map[name])
ax.set_xlabel('Epoch / Communication round')
ax.set_ylabel('Training loss')
ax.set_title('Training Loss Curves (all main models use step-decay LR)')
ax.grid(alpha=0.28)
ax.legend()
fig.tight_layout()
plt.show()
```



training_loss_curves.png

三模型混淆矩阵

集中式 MLP、FedAvg MLP 与 DP-FedAvg MLP 的混淆矩阵。

```
from pathlib import Path
import json
import numpy as np
import matplotlib.pyplot as plt

out = Path('data/mywork/final_outputs')
if not out.exists():
    out = Path.cwd()
# Plotting data comes from cached /train API response objects, not CSV summaries.
results = json.loads((out / 'experiment_results.json').read_text(encoding='utf-8'))
experiments = {item['name']: item for item in results['results']}

def metric_row(name):
    item = experiments[name]
    requested = item.get('requested_config') or {}
    dp = item.get('dp') or {}
    return {
        'name': name,
        **((item.get('metrics') or {})),
        'noise_multiplier': dp.get('noise_multiplier', requested.get('noise_multiplier')),
        'dirichlet_alpha': requested.get('dirichlet_alpha'),
    }

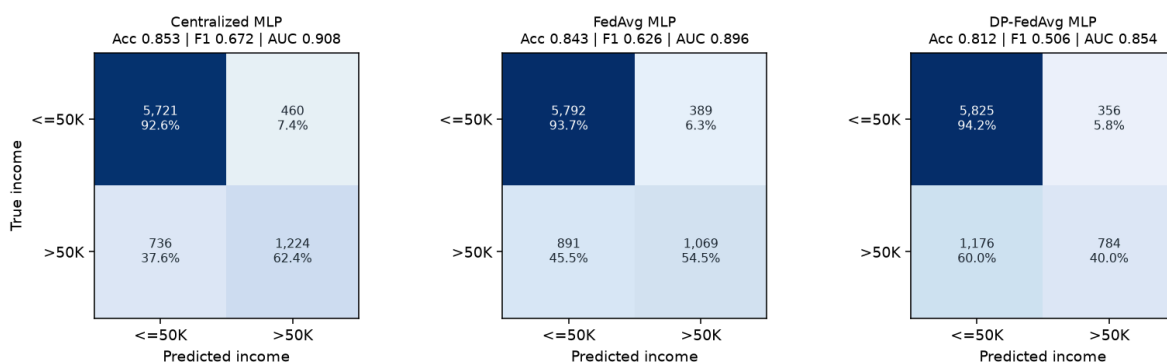
main_names = ['centralized_mlp', 'fedavg_mlp', 'dp_fedavg_mlp']
label_map = {
    'centralized_mlp': 'Centralized MLP',
    'fedavg_mlp': 'FedAvg MLP',
    'dp_fedavg_mlp': 'DP-FedAvg MLP',
}
```

```

class_labels = ['<=50K', '>50K']
matrices = {
    name: np.array(experiments[name]['metrics']['confusion_matrix'])
    for name in main_names
}
vmax = max(int(matrix.max()) for matrix in matrices.values())
fig, axes = plt.subplots(1, 3, figsize=(11.2, 4.2), dpi=140)
for idx, (ax, name) in enumerate(zip(axes, main_names)):
    matrix = matrices[name]
    metrics = experiments[name]['metrics']
    ax.imshow(matrix, cmap='Blues', vmin=0, vmax=vmax)
    ax.set_title(
        f"{label_map[name]}\n"
        f"Acc {metrics['accuracy']:.3f} | F1 {metrics['f1']:.3f} | AUC {metrics['auc']:.3f}"
        fontsize=9,
        fontweight='normal',
    )
    ax.set_xlabel('Predicted income')
    if idx == 0:
        ax.set_ylabel('True income')
    ax.set_xticks([0, 1], labels=class_labels)
    ax.set_yticks([0, 1], labels=class_labels)
    row_totals = matrix.sum(axis=1, keepdims=True)
    row_percentages = np.divide(
        matrix,
        row_totals,
        out=np.zeros_like(matrix, dtype=float),
        where=row_totals != 0,
    ) * 100
    for i in range(matrix.shape[0]):
        for j in range(matrix.shape[1]):
            color = 'white' if matrix[i, j] > vmax * 0.50 else '#1f2933'
            ax.text(
                j,
                i,
                f'{matrix[i, j]:,}\n{row_percentages[i, j]:.1f}%',
                ha='center',
                va='center',
                color=color,
                fontsize=8,
            )
fig.suptitle('Confusion Matrices (shared color scale)', y=0.96, fontsize=12)
fig.subplots_adjust(left=0.07, right=0.985, top=0.76, bottom=0.18, wspace=0.38)
plt.show()

```

Confusion Matrices (shared color scale)



confusion_matrices_main_models.png

DP 噪声敏感性

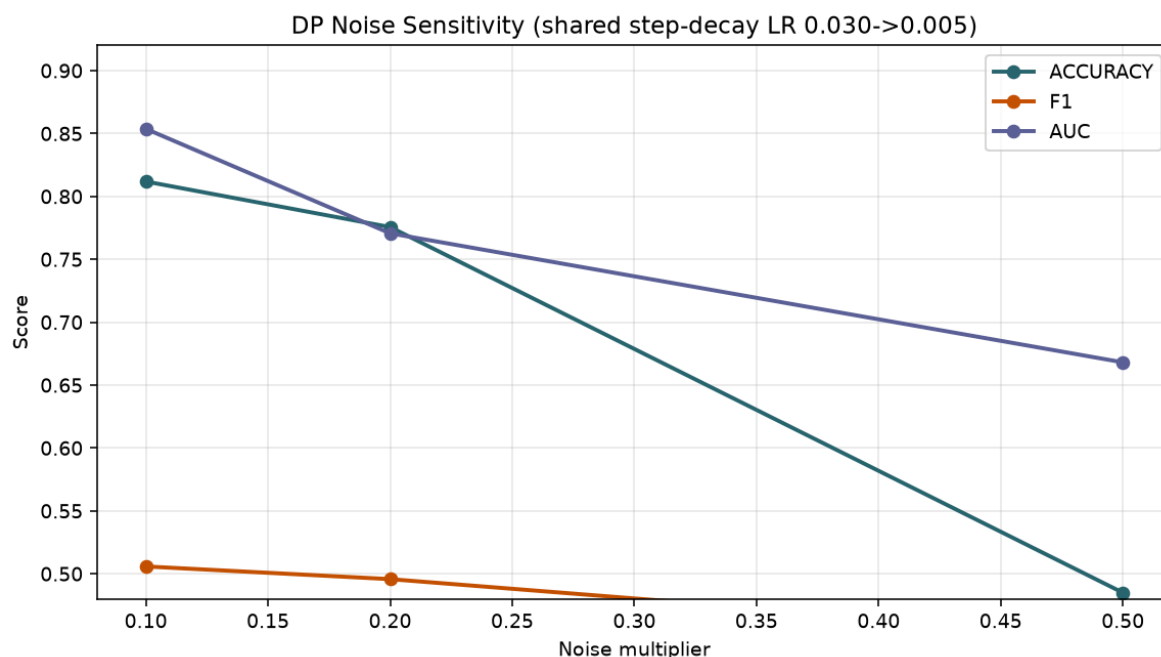
不同 noise_multiplier 下 DP-FedAvg 的指标变化，反映隐私扰动与效用之间的经验权衡。

```
from pathlib import Path
import json
import numpy as np
import matplotlib.pyplot as plt

out = Path('data/mywork/final_outputs')
if not out.exists():
    out = Path.cwd()
# Plotting data comes from cached /train API response objects, not CSV summaries.
results = json.loads((out / 'experiment_results.json').read_text(encoding='utf-8'))
experiments = {item['name']: item for item in results['results']}

def metric_row(name):
    item = experiments[name]
    requested = item.get('requested_config') or {}
    dp = item.get('dp') or {}
    return {
        'name': name,
        **(item.get('metrics') or {}),
        'noise_multiplier': dp.get('noise_multiplier', requested.get('noise_multiplier')),
        'dirichlet_alpha': requested.get('dirichlet_alpha'),
    }

noise_rows = [metric_row(name) for name in experiments if name.startswith('dp_noise_')]
noise_rows = sorted(noise_rows, key=lambda row: row['noise_multiplier'])
noise_values = [row['noise_multiplier'] for row in noise_rows]
lr_values = experiments['dp_noise_0.1']['history'].get('lr', [])
fig, ax = plt.subplots(figsize=(8.4, 4.8), dpi=140)
for metric, color in [('accuracy', '#28666e'), ('f1', '#c75000'), ('auc', '#5b5f97')]:
    ax.plot(noise_values, [row[metric] for row in noise_rows], marker='o', linewidth=2,
            color=color)
ax.set_xlabel('Noise multiplier')
ax.set_ylabel('Score')
ax.set_title(f'DP Noise Sensitivity (shared step-decay LR {lr_values[0]:.3f}->{lr_values[-1]:.3f})')
ax.set_ylim(0.48, 0.92)
ax.grid(alpha=0.28)
ax.legend()
fig.tight_layout()
plt.show()
```



dp_noise_sensitivity.png

alpha 敏感性

本实验使用显式客户端 CSV 固定划分，因此 alpha 主要是配置记录，不代表重新采样后的 Dirichlet Non-IID 强度。

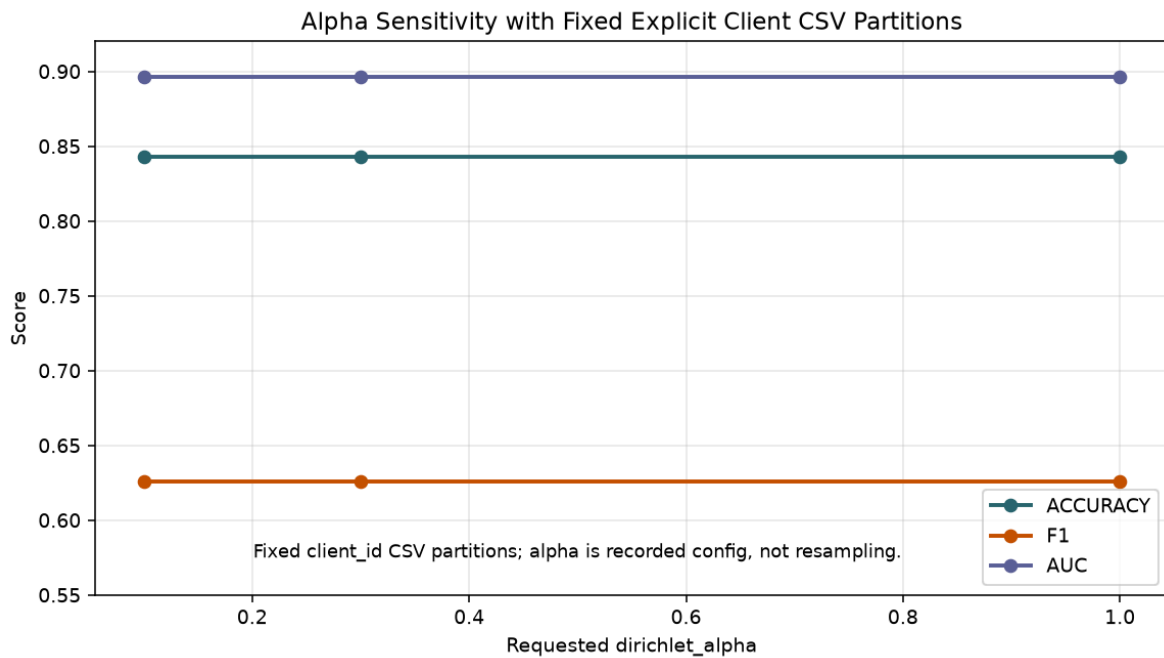
```
from pathlib import Path
import json
import numpy as np
import matplotlib.pyplot as plt

out = Path('data/mywork/final_outputs')
if not out.exists():
    out = Path.cwd()
# Plotting data comes from cached /train API response objects, not CSV summaries.
results = json.loads((out / 'experiment_results.json').read_text(encoding='utf-8'))
experiments = {item['name']: item for item in results['results']}

def metric_row(name):
    item = experiments[name]
    requested = item.get('requested_config') or {}
    dp = item.get('dp') or {}
    return {
        'name': name,
        **(item.get('metrics') or {}),
        'noise_multiplier': dp.get('noise_multiplier', requested.get('noise_multiplier')),
        'dirichlet_alpha': requested.get('dirichlet_alpha'),
    }

alpha_rows = [metric_row(name) for name in experiments if name.startswith('alpha_')]
alpha_rows = sorted(alpha_rows, key=lambda row: row['dirichlet_alpha'])
alpha_values = [row['dirichlet_alpha'] for row in alpha_rows]
fig, ax = plt.subplots(figsize=(8.4, 4.8), dpi=140)
```

```
for metric, color in [('accuracy', '#28666e'), ('f1', '#c75000'), ('auc', '#5b5f97')]:
    ax.plot(alpha_values, [row[metric] for row in alpha_rows], marker='o', linewidth=2,
ax.set_xlabel('Requested dirichlet_alpha')
ax.set_ylabel('Score')
ax.set_title('Alpha Sensitivity with Fixed Explicit Client CSV Partitions')
ax.set_ylim(0.55, 0.92)
ax.grid(alpha=0.28)
ax.legend()
ax.text(0.5, 0.575, 'Fixed client_id CSV partitions; alpha is recorded config, not resa
fig.tight_layout()
plt.show()
```



alpha_sensitivity.png

结论

集中式模型是合并数据参照；FedAvg 显示固定 Non-IID 客户端划分下的协作训练效果；DP-FedAvg 展示裁剪和高斯噪声带来的隐私-效用权衡。由于没有严格隐私 accountant，本实验不宣称 Opacus 级别的可证明隐私预算。